

MLOPS

Assignment#1

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Section - B

Github Repository

The student-ml-api GitHub repository contains the complete source code and MLOps configuration required for this assignment. The repository includes the FastAPI application, automated tests, Dockerfile, .dockerignore, VERSION file, and separate GitHub Actions workflows for Continuous Integration and automated releases. Development work was performed through feature branches and Pull Requests rather than directly on the protected main branch.

The screenshot shows the GitHub repository page for 'student-ml-api' by user 'SamiyaSaleem'. The repository is public and has 7 branches and 4 tags. The file list includes:

File	Commit Message	Time
github/workflows	feat: publish Docker image with commit SHA tag (#6)	3 hours ago
tests	feat: add model metadata for version 1.1.0 (#4)	16 hours ago
.dockerignore	Feat: Add prediction API and automated tests (#1)	19 hours ago
.gitignore	chore: initialize project repository	2 days ago
Dockerfile	feat: add OCI image metadata labels (#5)	3 hours ago
README.md	chore: initialize project repository	2 days ago
VERSION	feat: add model metadata for version 1.1.0 (#4)	16 hours ago
app.py	feat: add model metadata for version 1.1.0 (#4)	16 hours ago
pytest.ini	Feat: Add prediction API and automated tests (#1)	19 hours ago
requirements.txt	Feat: Add prediction API and automated tests (#1)	19 hours ago

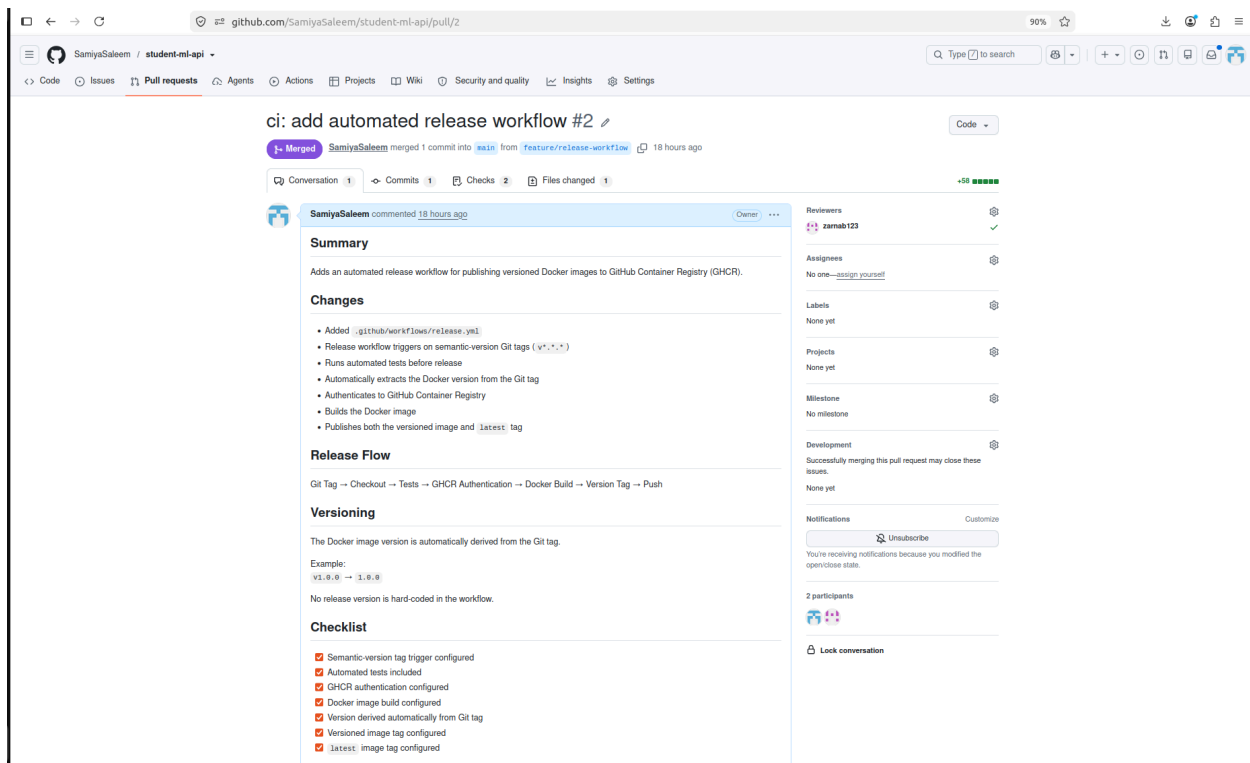
The README section is visible, titled 'student-ml-api', with the description: 'FastAPI prediction API for the Advanced MLOps Exercise.'

On the right sidebar, the 'About' section shows 'MLOps Assignment 1' with 0 stars, 0 watching, and 0 forks. The 'Releases' section shows 4 tags and a link to 'Create a new release'. The 'Packages' section shows 1 package, 'student-ml-api'. The 'Contributors' section shows 1 contributor, 'SamiyaSaleem'. The 'Languages' section shows a bar chart with Python at 71.2% and Dockerfile at 28.8%.

Professional Pull Requests

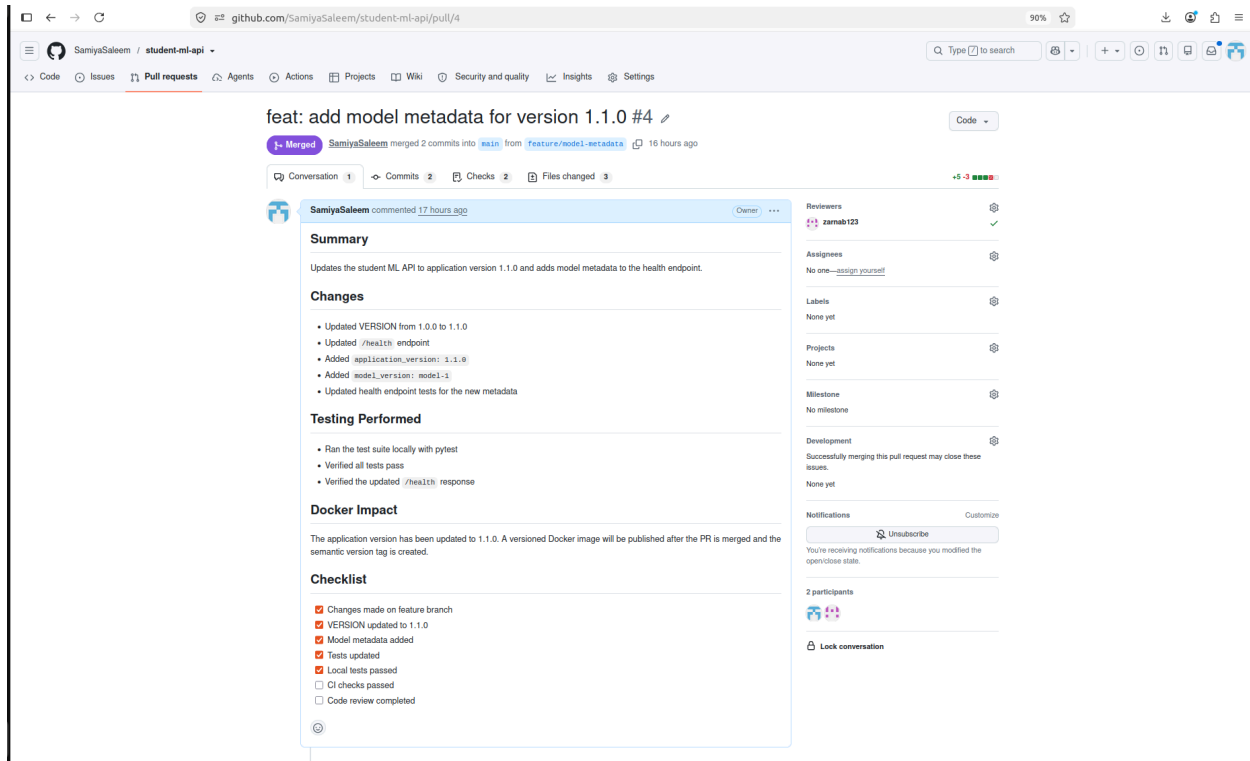
Pull Request – Automated Release Workflow

This Pull Request introduced the automated release pipeline for the project. The PR description clearly documents the purpose of the change, files and workflow changes, release flow, semantic versioning approach, and validation checklist. The workflow automatically derives the Docker image version from the Git tag and publishes versioned images to GitHub Container Registry after successful validation.



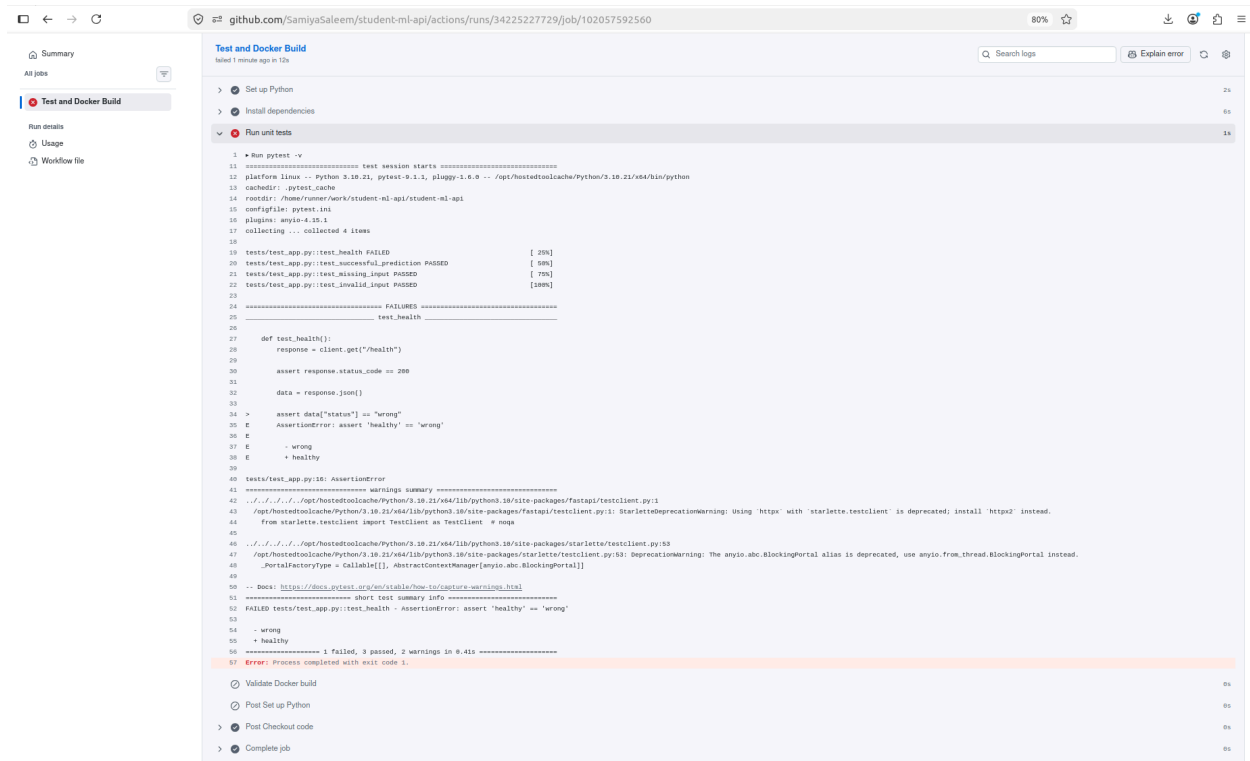
Pull Request – Model Metadata Version 1.1.0

This Pull Request introduced model metadata for application version 1.1.0. It updated the health endpoint and automated tests while following the required feature branch -> Pull Request -> CI -> review -> merge workflow. The PR documents the implemented changes, testing performed, Docker impact, and completion checklist.



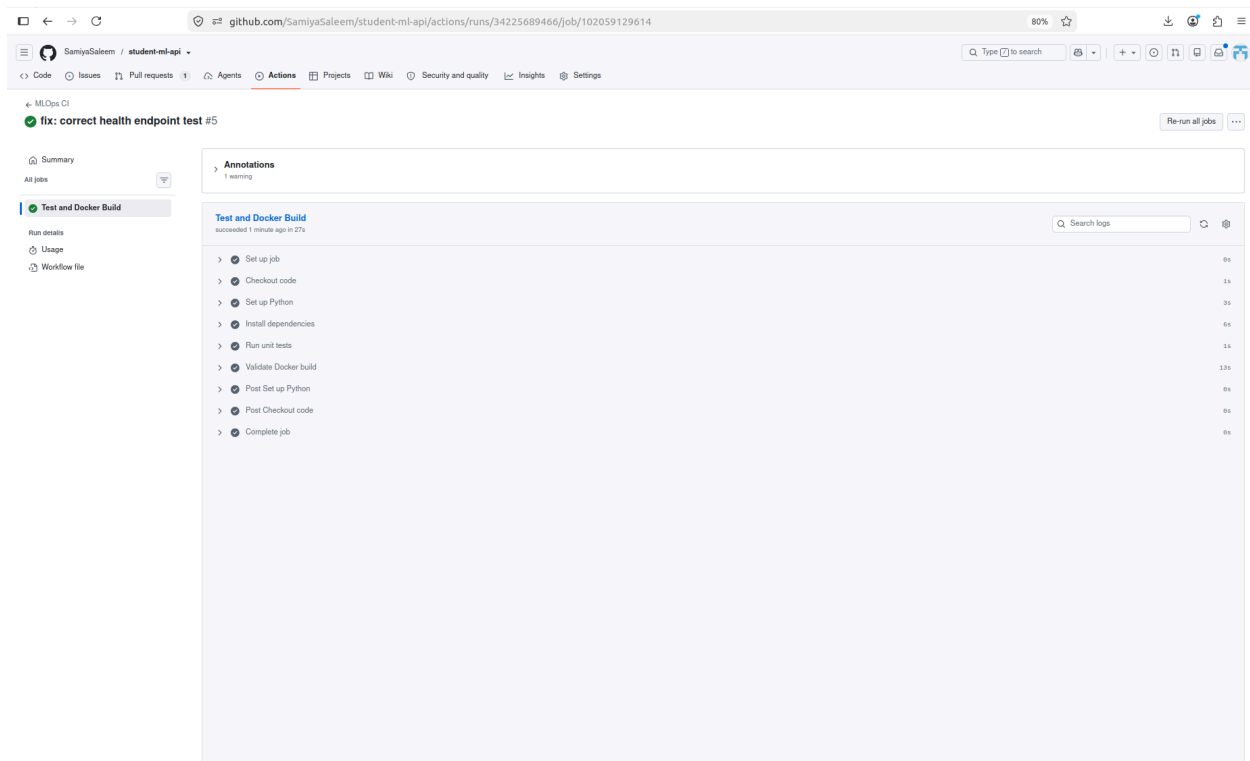
One failed CI Execution

This screenshot demonstrates an intentionally failed CI execution. A test assertion was deliberately changed so that the expected health status did not match the actual API response. pytest detected the failure, causing the GitHub Actions job to fail and preventing the change from being treated as valid. This demonstrates that the CI pipeline correctly detects test failures and acts as an automated quality gate.



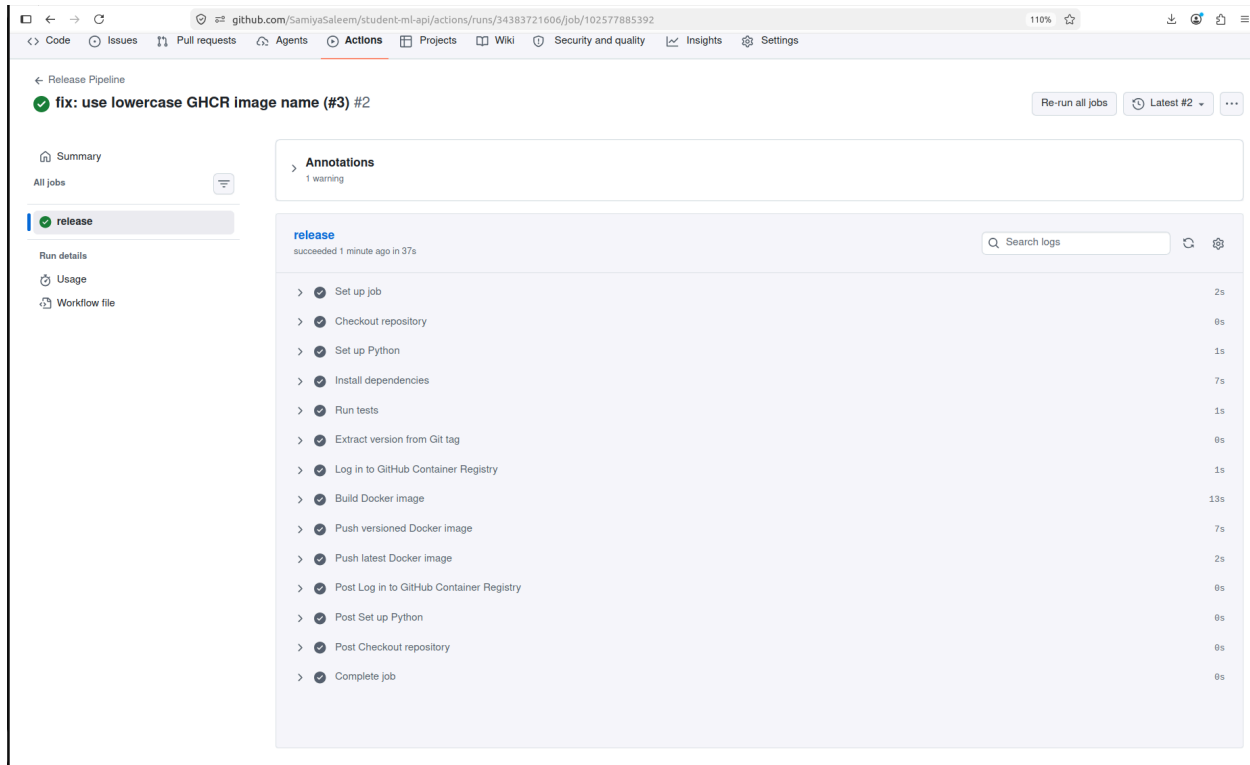
One successful CI execution

After correcting the intentionally broken test, the CI pipeline was executed again. All automated tests passed and the Docker build validation completed successfully. The green GitHub Actions result confirms that the application satisfied the configured CI quality checks.



One successful release execution

This screenshot demonstrates a successful automated release pipeline. The release workflow checked out the repository, installed dependencies, executed automated tests, extracted the application version from the Git tag, authenticated with GHCR, built the Docker image, and published the required Docker image tags. This confirms that Docker images are published through automation rather than being manually uploaded.



Registry Evidence

GitHub Container Registry (GHCR) stores the versioned Docker artifacts produced by the release workflow. The registry contains the required 1.0.0 and 1.1.0 images together with the latest tag. Keeping explicit version tags allows previous releases to remain available for reproducible deployments and rollback.

The screenshot shows the Docker Hub page for the 'student-ml-api' image, version 1.1.0. The page is titled 'student-ml-api 1.1.0' and includes a 'Private' badge. The main content area displays the 'Install from the command line' section with the command: `$ docker pull ghcr.io/samiyasaleem/student-ml-api:1.1.0`. Below this, the 'Recent tagged image versions' section lists three versions: 1.1.0 (latest), 1.0.1, and 1.0.0. The 'Details' sidebar on the right shows the repository name 'student-ml-api', the owner 'SamiyaSaleem', and the last published time '1 minute ago'. The 'README.md' section at the bottom describes the image as a 'FastAPI prediction API for the Advanced MLOps Exercise'.

Release Tags

Git tags provide the source-control version associated with each release. The repository contains the required v1.0.0 and v1.1.0 tags, along with additional patch-release tags created during later workflow improvements. These tags provide traceability between Git history and the corresponding Docker releases.

The screenshot shows the GitHub repository page for 'student-ml-api' with the 'Tags' tab selected. The page displays a list of four tags: v1.1.1, v1.1.0, v1.0.1, and v1.0.0. Each tag entry includes the version number, a 'Verified' badge, and a list of download links for the corresponding Docker image. The 'v1.1.1' tag is the most recent, published 3 hours ago. The 'v1.1.0' tag was published 17 hours ago. The 'v1.0.1' tag was published 17 hours ago. The 'v1.0.0' tag was published 18 hours ago.

Demonstration

The following evidence demonstrates the complete reproducible deployment workflow. The repository is cloned from GitHub, its development history and Pull Requests are inspected, GitHub Actions executions are reviewed, a versioned Docker image is pulled from GHCR and executed, the API is tested, and finally the application is rolled back to a previous stable version.

Clone repository

The repository was cloned directly from GitHub using git clone. The successful clone confirms that the complete project can be obtained from version control on a clean working directory.

```
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop$ git clone http://github.com/SamiyaSaleem/student-ml-api.git
Cloning into 'student-ml-api'...
warning: redirecting to https://github.com/SamiyaSaleem/student-ml-api.git/
remote: Enumerating objects: 60, done.
remote: Counting objects: 100% (60/60), done.
remote: Compressing objects: 100% (39/39), done.
remote: Total 60 (delta 18), reused 47 (delta 11), pack-reused 0 (from 0)
Receiving objects: 100% (60/60), 11.41 KiB | 11.41 MiB/s, done.
Resolving deltas: 100% (18/18), done.
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop$
```

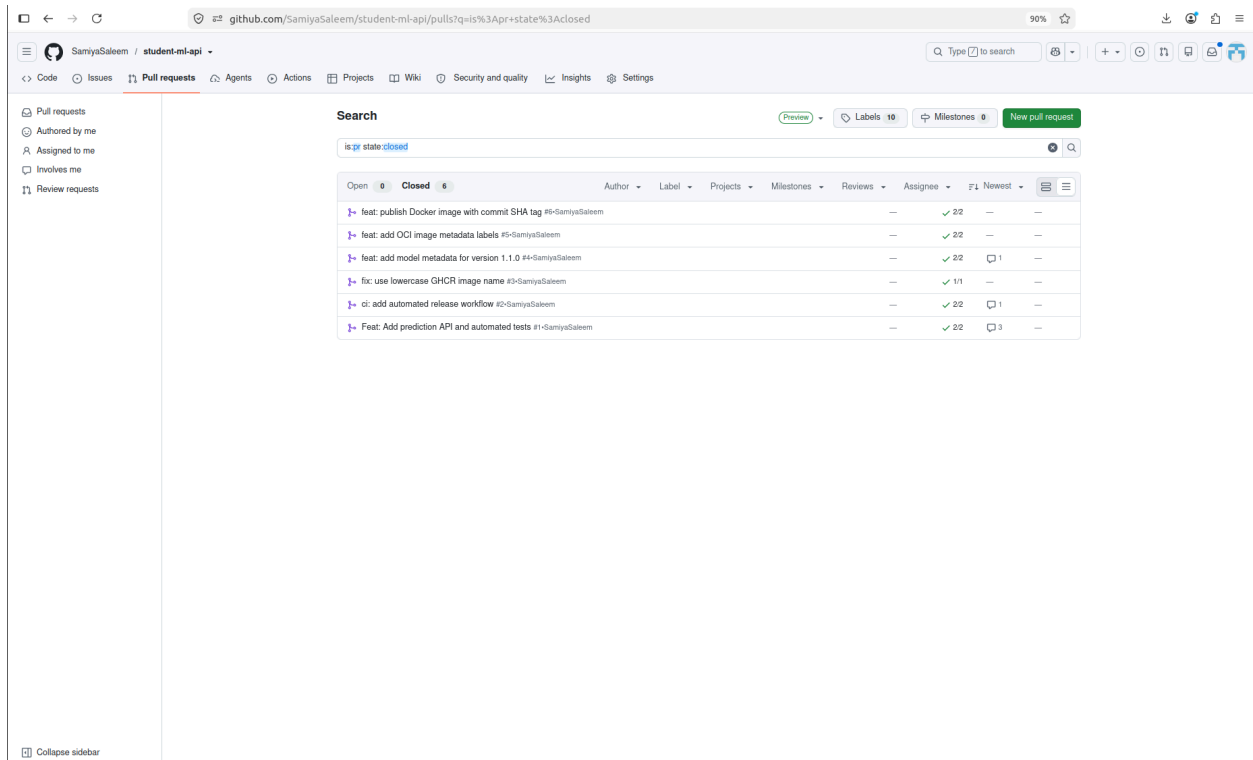
Inspect Git History

The Git history was inspected using git log --oneline --decorate --graph, showing the sequence of merged feature work and release-related commits. The Git tags were also listed, confirming the availability of multiple versioned releases including v1.0.0 and v1.1.0.

```
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop$ cd student-ml-api
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop/student-ml-api$ ^[[200~git log --oneline --decorate --graph -10~
git: command not found
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop/student-ml-api$ git log --oneline --decorate --graph -10
* 3d275cd (HEAD -> main, tag: v1.1.1, origin/main, origin/HEAD) feat: publish Docker image with commit SHA tag (#6)
* 84700ae feat: add OCI image metadata labels (#5)
* da51780 (tag: v1.1.0) feat: add model metadata for version 1.1.0 (#4)
* 2a88098 (tag: v1.0.1) fix: use lowercase GHCR image name (#3)
* 764698e (tag: v1.0.0) ci: add automated release workflow (#2)
* ada72c8 Feat: Add prediction API and automated tests (#1)
* a4bdca5 chore: initialize project repository
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop/student-ml-api$ git tag --list
v1.0.0
v1.0.1
v1.1.0
v1.1.1
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop/student-ml-api$
```

Inspect PR

The closed Pull Requests were inspected on GitHub. The PR history demonstrates that development changes were introduced through branches and Pull Requests rather than being developed directly on main. Multiple PRs are visible for the prediction API, release automation, GHCR correction, model metadata, OCI metadata, and commit-SHA image tagging.



Search

is pr state: closed

Open	Closed	Author	Label	Projects	Milestones	Reviews	Assignee	File	Newest
	6								
		feat: publish Docker image with commit SHA tag #9-SamiyaSaleem				2/2			
		feat: add OCI image metadata labels #9-SamiyaSaleem				2/2			
		feat: add model metadata for version 1.1.0 #4-SamiyaSaleem				2/2		1	
		fix: use lowercase GHCR image name #3-SamiyaSaleem				1/1			
		ci: add automated release workflow #2-SamiyaSaleem				2/2		1	
		Feat: Add prediction API and automated tests #1-SamiyaSaleem				2/2		3	

Inspect Github Action

The GitHub Actions history was inspected to verify automated CI and release executions. The repository contains separate MLOps CI and Release Pipeline workflows, demonstrating separation between code validation and artifact publishing.

Actions New workflow

All workflows Filter workflow runs

Showing runs from all workflows

Help us improve GitHub Actions
Tell us how to make GitHub Actions work better for you with three quick questions. Give feedback

Workflow	Event	Status	Branch	Actor
feat: publish Docker image with commit SHA tag (#6)	Release Pipeline #4: Commit 3d275cd pushed by SamiyaSaleem	Success	feature/commit-sha-tag	Today at 12:48 PM
feat: publish Docker image with commit SHA tag	MLOps CI #15: Pull request #5 opened by SamiyaSaleem	Success	feature/commit-sha-tag	Today at 12:45 PM
feat: publish Docker image with commit SHA tag	MLOps CI #14: Commit a70eeb pushed by SamiyaSaleem	Success	feature/commit-sha-tag	Today at 12:44 PM
feat: add OCI image metadata labels	MLOps CI #13: Pull request #5 opened by SamiyaSaleem	Success	feature/image-metadata	Today at 12:27 PM
feat: add OCI image metadata labels	MLOps CI #12: Commit b49d8d4 pushed by SamiyaSaleem	Success	feature/image-metadata	Today at 12:25 PM
feat: add model metadata for version 1.1.0 (#4)	Release Pipeline #3: Commit da51780 pushed by SamiyaSaleem	Success	feature/model-metadata	Sep 9, 11:35 PM GMT+5
feat: add model metadata for version 1.1.0	MLOps CI #11: Pull request #4 opened by SamiyaSaleem	Success	feature/model-metadata	Sep 9, 11:19 PM GMT+5
test: update health endpoint tests for model metadata	MLOps CI #10: Commit b8979d pushed by SamiyaSaleem	Success	feature/model-metadata	Sep 9, 11:17 PM GMT+5
fix: use lowercase GHCR image name (#3)	Release Pipeline #2: Commit 2a8809b pushed by SamiyaSaleem	Success	fix/lowercase-ghcr	Sep 9, 10:34 PM GMT+5
fix: use lowercase GHCR image name	MLOps CI #9: Pull request #3 opened by SamiyaSaleem	Success	fix/lowercase-ghcr	Sep 9, 10:29 PM GMT+5
ci: add automated release workflow (#2)	Release Pipeline #1: Commit 764888e pushed by SamiyaSaleem	Success	feature/model-metadata	Sep 9, 10:17 PM GMT+5

Pull image from registry

The versioned Docker image `ghcr.io/samiyasaleem/student-ml-api:1.1.0` was pulled directly from GHCR. Docker confirmed the image digest and successful availability of the artifact locally. This demonstrates that deployment can use the previously published registry artifact without rebuilding the application from source.

```
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop/student-ml-api$ docker pull ghcr.io/samiyasaleem/student-ml-api:1.1.0
1.1.0: Pulling from samiyasaleem/student-ml-api
Digest: sha256:c62b24c72834db64053720b111d339a6b52c1cebfaa6151e47ab0c416a65dff3
Status: Image is up to date for ghcr.io/samiyasaleem/student-ml-api:1.1.0
ghcr.io/samiyasaleem/student-ml-api:1.1.0
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop/student-ml-api$ docker images ghcr.io/samiyasaleem/student-ml-api
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
ghcr.io/samiyasaleem/student-ml-api:1.0.0	7866ea53e952	244MB	59.3MB	
ghcr.io/samiyasaleem/student-ml-api:1.1.0	c62b24c72834	245MB	58.9MB	
ghcr.io/samiyasaleem/student-ml-api:1.1.1	c26c9d2acd19	245MB	58.9MB	
ghcr.io/samiyasaleem/student-ml-api:3d275cd	c26c9d2acd19	245MB	58.9MB	
ghcr.io/samiyasaleem/student-ml-api:latest	c26c9d2acd19	245MB	58.9MB	

```
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop/student-ml-api$
```

Run Container

The downloaded 1.1.0 Docker image was started as a container with host port 5000 mapped to container port 5000. docker ps confirmed that the container was running successfully and exposing the expected API port.

```
user@user-ideaPad-Flex-5-14ALC7: ~/Desktop/student-ml-api
(.venv) user@user-ideaPad-Flex-5-14ALC7:~/Desktop/student-ml-api$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
9ed291487895   ghcr.io/samiyasaleem/mlops-cd-demo:1.0.0 "python app.py"         6 hours ago   Up 6 hours   0.0.0.0:5001->5001/tcp, [::]:5001->5001/tcp
p_mlops-api
(.venv) user@user-ideaPad-Flex-5-14ALC7:~/Desktop/student-ml-api$ docker run -d \
--name demo-student-ml-api \
-p 5000:5000 \
ghcr.io/samiyasaleem/student-ml-api:1.1.0
48fc5eb2d27be54de4b66580e4532d2f92d10ae2f8e73b7288dc0679216249e1
(.venv) user@user-ideaPad-Flex-5-14ALC7:~/Desktop/student-ml-api$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
48fc5eb2d27b   ghcr.io/samiyasaleem/student-ml-api:1.1.0 "uvicorn app:app --h..." 9 seconds ago   Up 8 seconds   0.0.0.0:5000->5000/tcp, [::]:5
000->5000/tcp
demo-student-ml-api
9ed291487895   ghcr.io/samiyasaleem/mlops-cd-demo:1.0.0 "python app.py"         6 hours ago   Up 6 hours   0.0.0.0:5001->5001/tcp, [::]:5
001->5001/tcp
mlops-api
(.venv) user@user-ideaPad-Flex-5-14ALC7:~/Desktop/student-ml-api$
```

Test API

The running container was validated through both API endpoints. GET /health returned a healthy status with application version 1.1.0 and model version model-1. POST /predict with an input value of 10 returned the expected prediction of 20.0. This confirms that the published container artifact operates correctly after deployment.

```
(.venv) user@user-ideaPad-Flex-5-14ALC7:~/Desktop/student-ml-api$ curl http://localhost:5000/health
{"status": "healthy", "application": "student-ml-api", "application_version": "1.1.0", "model_version": "model-1"}(.venv) user@user-ideaPad-Flex-5-14ALC7:~/Desktop/student-ml-api$ curl -X POST http://localhost:5000/predict \
-H "Content-Type: application/json" \
-d '{"value":10}'
{"input":10.0,"prediction":20.0}(.venv) user@user-ideaPad-Flex-5-14ALC7:~/Desktop/student-ml-api$
```

Rollback

A rollback was performed by stopping the newer 1.1.0 deployment and running the previously published 1.0.0 Docker image from GHCR. No source-code modification, dependency installation, or Docker rebuild was required. The /health endpoint returned version 1.0.0, confirming that the application had successfully been restored to the previous stable release. This demonstrates the reproducibility and operational value of retaining immutable versioned artifacts in the container registry.

```
user@user-ideaPad-Flex-5-14ALC7: ~/Desktop/mlops-A1/student-ml-api
5a62435fe308: Pull complete
5f10ca603e75: Pull complete
Digest: sha256:c62b24c72834db64053720b111d339a6b52c1cebfaa6151e47ab0c416a65dff3
Status: Downloaded newer image for ghcr.io/samiyasaleem/student-ml-api:1.1.0
ghcr.io/samiyasaleem/student-ml-api:1.1.0
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop/mlops-A1/student-ml-api$ docker pull ghcr.io/samiyasaleem/student-ml-api:latest
latest: Pulling from samiyasaleem/student-ml-api
Digest: sha256:c62b24c72834db64053720b111d339a6b52c1cebfaa6151e47ab0c416a65dff3
Status: Downloaded newer image for ghcr.io/samiyasaleem/student-ml-api:latest
ghcr.io/samiyasaleem/student-ml-api:latest
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop/mlops-A1/student-ml-api$ docker rm -f student-ml-api
Error response from daemon: No such container: student-ml-api
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop/mlops-A1/student-ml-api$ docker run -d \
  --name student-ml-api \
  -p 5000:5000 \
  ghcr.io/samiyasaleem/student-ml-api:1.0.0
257da3a7f535
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop/mlops-A1/student-ml-api$ docker ps
CONTAINER ID   IMAGE                                     COMMAND                  CREATED        STATUS        PORTS
257da3a7f535   ghcr.io/samiyasaleem/student-ml-api:1.0.0   "uvicorn app:app --h..."   8 seconds ago   Up 7 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop/mlops-A1/student-ml-api$ curl http://localhost:5000/health
{"status": "healthy", "application": "student-ml-api", "version": "1.0.0"}
(.venv) user@user-IdeaPad-Flex-5-14ALC7:~/Desktop/mlops-A1/student-ml-api$
```